

Coil Sensitivity Analysis

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1 Background: Magnetic Field Modeling

Consider an array of 4×4 electromagnetic sub-coils. Each sub-coil (i, j) is located at position $\mathbf{p}_{i,j} \in \mathbb{R}^3$ with orientation unit vector $\mathbf{n}_{i,j} \in \mathbb{S}^2$, as depicted in Figure 1. Each coil is driven by an AC current $I_{i,j}(t)$, defined by:

$$I_{i,j}(t) = I_0 \cos(\omega t + \psi_{i,j}), \quad (1)$$

where I_0 is the peak AC current amplitude, ω is the drive angular frequency, and $\psi_{i,j}$ is the relative actuation phase shift of coil (i, j) .

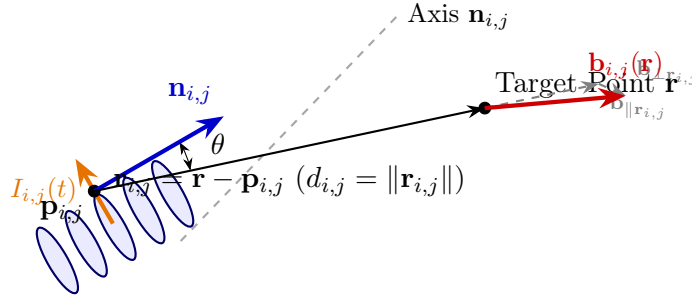


Figure 1: Geometry of a single sub-coil (i, j) located at position $\mathbf{p}_{i,j}$ with orientation unit vector $\mathbf{n}_{i,j}$ driven by time-varying current $I_{i,j}(t)$, generating magnetic field response $\mathbf{b}_{i,j}(\mathbf{r})$ at target point $\mathbf{r}$.

Under the point-dipole approximation, the magnetic flux density response per unit current generated by sub-coil (i, j) at target point $\mathbf{r} \in \mathbb{R}^3$ is given by:

$$\mathbf{b}_{i,j}(\mathbf{r}) = K_{\text{dipole}} \left(\frac{3(\mathbf{r}_{i,j} \cdot \mathbf{n}_{i,j})\mathbf{r}_{i,j}}{d_{i,j}^5} - \frac{\mathbf{n}_{i,j}}{d_{i,j}^3} \right),$$

where $\mathbf{r}_{i,j} = \mathbf{r} - \mathbf{p}_{i,j}$ is the relative position vector, $d_{i,j} = \|\mathbf{r}_{i,j}\|$ is the Euclidean distance, and $K_{\text{dipole}} = \frac{\mu_0 m_0}{4\pi}$ represents the unit-current magnetic dipole strength prefactor ($m_0 = NA$ for a coil with N turns and effective cross-sectional area A).

Applying linear superposition across all 16 driven coils and expanding $I_{i,j}(t)$ using the cosine angle addition identity gives the total time-varying field $\mathbf{B}(\mathbf{r}, t)$:

$$\mathbf{B}(\mathbf{r}, t) = \sum_{i=1}^4 \sum_{j=1}^4 \mathbf{b}_{i,j}(\mathbf{r}) I_{i,j}(t) = \mathbf{u}(\mathbf{r}) \cos(\omega t) + \mathbf{v}(\mathbf{r}) \sin(\omega t),$$

where the spatial field is resolved into orthogonal in-phase $\mathbf{u}(\mathbf{r})$ and 90°-lagged (quadrature) $\mathbf{v}(\mathbf{r})$ vector components:

$$\mathbf{u}(\mathbf{r}) = I_0 \sum_{i=1}^4 \sum_{j=1}^4 \mathbf{b}_{i,j}(\mathbf{r}) \cos \psi_{i,j}, \quad \mathbf{v}(\mathbf{r}) = -I_0 \sum_{i=1}^4 \sum_{j=1}^4 \mathbf{b}_{i,j}(\mathbf{r}) \sin \psi_{i,j}.$$

From $\mathbf{u}(\mathbf{r})$ and $\mathbf{v}(\mathbf{r})$, the field rotation vector $\mathbf{w}(\mathbf{r})$, local rotation axis $\mathbf{n}_{\text{rot}}(\mathbf{r})$, root-mean-square (RMS) field magnitude $B_{\text{rms}}(\mathbf{r})$, and lateral magnetic force driver gradient $\mathbf{g}_{\text{lat}}(\mathbf{r})$ are computed as:

$$\mathbf{w}(\mathbf{r}) = \mathbf{u}(\mathbf{r}) \times \mathbf{v}(\mathbf{r}), \quad \mathbf{n}_{\text{rot}}(\mathbf{r}) = \frac{\mathbf{w}(\mathbf{r})}{\|\mathbf{w}(\mathbf{r})\|},$$

$$B_{\text{rms}}(\mathbf{r}) = \sqrt{\frac{1}{2} (\|\mathbf{u}(\mathbf{r})\|^2 + \|\mathbf{v}(\mathbf{r})\|^2)}, \quad \mathbf{g}_{\text{lat}}(\mathbf{r}) = \nabla_{xy} (B_{\text{rms}}^2(\mathbf{r})).$$

2 Frames and Kinematics in $\mathfrak{se}(3)$

We define frames and sub-coil poses using homogeneous transformation matrices $T_{i,j} \in SE(3)$, parameterized by rotation matrix $R_{i,j} \in SO(3)$ (where $\mathbf{n}_{i,j} = R_{i,j}\mathbf{e}_z$) and translation vector $\mathbf{p}_{i,j} \in \mathbb{R}^3$:

$$T_{i,j} = \begin{bmatrix} R_{i,j} & \mathbf{p}_{i,j} \\ \mathbf{0}^T & 1 \end{bmatrix}.$$

When performing operations across frame transformations, matrix compositions combine both components directly:

$$T_{W \leftarrow C_i} = \text{Rot}_Z(\Phi_i) T_{W \leftarrow C_1}, \quad \mathbf{p}_{i,j} = R_i \mathbf{p}_j^{(t)}, \quad \mathbf{n}_{i,j} = R_i \mathbf{n}_j^{(t)}.$$

3 Additive Gaussian Pose Noise

For each coil pose $T_{i,j}$, we apply additive zero-mean Gaussian noise $\boldsymbol{\xi}_{i,j} = [\boldsymbol{\omega}_{i,j}^T, \boldsymbol{\nu}_{i,j}^T]^T \in \mathfrak{se}(3)$:

$$\boldsymbol{\xi}_{i,j} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\xi), \quad \boldsymbol{\Sigma}_\xi = \text{diag}(\sigma_{\text{rot}}^2 \mathbf{I}_3, \sigma_{\text{trans}}^2 \mathbf{I}_3).$$

To first order, position perturbations $\delta \mathbf{p}_{i,j}$ and orientation perturbations $\delta \mathbf{n}_{i,j}$ follow:

$$\delta \mathbf{p}_{i,j} = \boldsymbol{\nu}_{i,j}, \quad \delta \mathbf{n}_{i,j} = \boldsymbol{\omega}_{i,j} \times \mathbf{n}_{i,j} = -\hat{\mathbf{n}}_{i,j} \boldsymbol{\omega}_{i,j},$$

where $\hat{\mathbf{a}}$ denotes the 3×3 skew-symmetric cross-product matrix of vector $\mathbf{a}$:

$$\hat{\mathbf{a}} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}.$$

4 Objective Function Formulation

We formulate general cost terms integrated over a 2D/3D region of interest Ω :

$$\mathcal{J}_{\text{sens}} = \int_{\Omega} \sum_{i=1}^4 \sum_{j=1}^4 \text{Tr} \left(\mathbf{J}_{i,j}^{\mathbf{n}_{\text{rot}}}(\mathbf{r}) \boldsymbol{\Sigma}_\xi \left(\mathbf{J}_{i,j}^{\mathbf{n}_{\text{rot}}}(\mathbf{r}) \right)^T \right) d\mathbf{r},$$

$$\mathcal{J}_{\text{mag}} = - \int_{\Omega} B_{\text{rms}}(\mathbf{r}) d\mathbf{r}, \quad \mathcal{J}_{\text{grad}} = \int_{\Omega} \|\mathbf{g}_{\text{lat}}(\mathbf{r})\|^2 d\mathbf{r}.$$

We investigate coupled coil plates versus individually movable coil designs, and model perturbations individually as follows:

- **Coupled Coil Plates:** Mechanical noise cascades through the assembly. Variance is added at the cluster baseplate level ($\boldsymbol{\Sigma}_{\text{plate}}$) as well as the individual sub-coil mounts ($\boldsymbol{\Sigma}_{\text{coil}}$), doubling total variance ($\boldsymbol{\Sigma}_\xi = \boldsymbol{\Sigma}_{\text{plate}} + \boldsymbol{\Sigma}_{\text{coil}} \approx 2\boldsymbol{\Sigma}_{\text{coil}}$).
- **Individually Mounted Coils:** Each sub-coil is positioned independently, resulting in single-level noise ($\boldsymbol{\Sigma}_\xi = \boldsymbol{\Sigma}_{\text{coil}}$).

5 Perturbation Propagation

To implement this sensitivity framework in symbolic optimization tools like CasADi, we physically derive the exact closed-form Jacobians mapping $\mathfrak{se}(3)$ perturbations $\xi_{i,j}$ to the rotation axis variation $\delta \mathbf{n}_{\text{rot}}(\mathbf{r})$.

5.1 Physical Derivation of Single-Coil Flux Density Sensitivity

Consider the dipole response $\mathbf{b}_{i,j}(\mathbf{r})$. Taking the total differential with respect to position vector $\mathbf{p}_{i,j}$ and unit normal vector $\mathbf{n}_{i,j}$ gives:

$$\delta \mathbf{b}_{i,j} = \frac{\partial \mathbf{b}_{i,j}}{\partial \mathbf{p}_{i,j}} \delta \mathbf{p}_{i,j} + \frac{\partial \mathbf{b}_{i,j}}{\partial \mathbf{n}_{i,j}} \delta \mathbf{n}_{i,j}.$$

1. Position Derivative Matrix ($\mathbf{J}_{\mathbf{p}_{i,j}}^{\mathbf{b}} \in \mathbb{R}^{3 \times 3}$): Since $\mathbf{r}_{i,j} = \mathbf{r} - \mathbf{p}_{i,j}$, taking spatial gradients with respect to $\mathbf{p}_{i,j}$ yields $\frac{\partial \mathbf{r}_{i,j}}{\partial \mathbf{p}_{i,j}} = -\mathbf{I}_3$ and $\frac{\partial d_{i,j}}{\partial \mathbf{p}_{i,j}} = -\frac{\mathbf{r}_{i,j}^T}{d_{i,j}}$. Differentiating $\mathbf{b}_{i,j}$ using product and chain rules yields:

$$\mathbf{J}_{\mathbf{p}_{i,j}}^{\mathbf{b}} = \frac{\partial \mathbf{b}_{i,j}}{\partial \mathbf{p}_{i,j}} = -K_{\text{dipole}} \left[\frac{3(\mathbf{r}_{i,j}^T \mathbf{n}_{i,j}) \mathbf{I}_3 + 3\mathbf{r}_{i,j} \mathbf{n}_{i,j}^T + 3\mathbf{n}_{i,j} \mathbf{r}_{i,j}^T}{d_{i,j}^5} - \frac{15(\mathbf{r}_{i,j}^T \mathbf{n}_{i,j}) \mathbf{r}_{i,j} \mathbf{r}_{i,j}^T}{d_{i,j}^7} \right].$$

- $d_{i,j}^{-5}$: Accounts for second-order spatial decay of the magnetic gradient tensor.
- $d_{i,j}^{-7}$: Captures higher-order geometric dilation from distance perturbations along vector $\mathbf{r}_{i,j}$.

2. Orientation Derivative Matrix ($\mathbf{J}_{\mathbf{n}_{i,j}}^{\mathbf{b}} \in \mathbb{R}^{3 \times 3}$): Differentiating $\mathbf{b}_{i,j}$ with respect to coil normal vector $\mathbf{n}_{i,j}$ gives:

$$\frac{\partial \mathbf{b}_{i,j}}{\partial \mathbf{n}_{i,j}} = K_{\text{dipole}} \left[\frac{3\mathbf{r}_{i,j} \mathbf{r}_{i,j}^T}{d_{i,j}^5} - \frac{\mathbf{I}_3}{d_{i,j}^3} \right].$$

Substituting the differential rotation kinematic relationship $\delta \mathbf{n}_{i,j} = -\hat{\mathbf{n}}_{i,j} \boldsymbol{\omega}_{i,j}$ into the total differential yields:

$$\mathbf{J}_{\boldsymbol{\omega}_{i,j}}^{\mathbf{b}} = \frac{\partial \mathbf{b}_{i,j}}{\partial \mathbf{n}_{i,j}} \frac{\partial \mathbf{n}_{i,j}}{\partial \boldsymbol{\omega}_{i,j}} = -K_{\text{dipole}} \left[\frac{3\mathbf{r}_{i,j} \mathbf{r}_{i,j}^T}{d_{i,j}^5} - \frac{\mathbf{I}_3}{d_{i,j}^3} \right] \hat{\mathbf{n}}_{i,j}.$$

Concatenating both blocks forms the block flux density Jacobian $\mathbf{J}_{i,j}^{\mathbf{b}} = [\mathbf{J}_{\boldsymbol{\omega}_{i,j}}^{\mathbf{b}} \quad \mathbf{J}_{\mathbf{p}_{i,j}}^{\mathbf{b}}] \in \mathbb{R}^{3 \times 6}$ such that $\delta \mathbf{b}_{i,j} = \mathbf{J}_{i,j}^{\mathbf{b}} \xi_{i,j}$.

5.2 Propagation to Time-Separated Field Vectors

Because the total field is a linear combination of individual sub-coil fields weighted by current phase $\psi_{i,j}$, the variations in in-phase vector $\mathbf{u}(\mathbf{r})$ and lagging vector $\mathbf{v}(\mathbf{r})$ with respect to pose noise $\xi_{i,j}$ are:

$$\frac{\partial \mathbf{u}}{\partial \xi_{i,j}} = I_0 \cos \psi_{i,j} \mathbf{J}_{i,j}^{\mathbf{b}}, \quad \frac{\partial \mathbf{v}}{\partial \xi_{i,j}} = -I_0 \sin \psi_{i,j} \mathbf{J}_{i,j}^{\mathbf{b}}.$$

5.3 Cross-Product Propagation to Rotation Vector $\mathbf{w}(\mathbf{r})$

The unnormalized rotation vector is defined as $\mathbf{w}(\mathbf{r}) = \mathbf{u}(\mathbf{r}) \times \mathbf{v}(\mathbf{r})$. Differentiating via the vector product rule gives:

$$\delta \mathbf{w} = \delta \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \delta \mathbf{v} = -\mathbf{v} \times \delta \mathbf{u} + \mathbf{u} \times \delta \mathbf{v}.$$

Converting cross products to skew-symmetric matrix multiplication ($-\mathbf{v} \times \delta \mathbf{u} = -\hat{\mathbf{V}} \delta \mathbf{u}$ and $\mathbf{u} \times \delta \mathbf{v} = \hat{\mathbf{U}} \delta \mathbf{v}$) yields the Jacobian $\mathbf{J}_{i,j}^{\mathbf{w}} = \frac{\partial \mathbf{w}}{\partial \xi_{i,j}} \in \mathbb{R}^{3 \times 6}$:

$$\mathbf{J}_{i,j}^{\mathbf{w}} = -\hat{\mathbf{V}} \left(I_0 \cos \psi_{i,j} \mathbf{J}_{i,j}^{\mathbf{b}} \right) - \hat{\mathbf{U}} \left(I_0 \sin \psi_{i,j} \mathbf{J}_{i,j}^{\mathbf{b}} \right).$$

- $\hat{\mathbf{U}}, \hat{\mathbf{V}} \in \mathbb{R}^{3 \times 3}$: Skew-symmetric matrices constructed from field vectors $\mathbf{u}(\mathbf{r})$ and $\mathbf{v}(\mathbf{r})$.

5.4 Projection to Unit Rotation Axis $\mathbf{n}_{\text{rot}}(\mathbf{r})$

Differentiating the normalization operator $\mathbf{n}_{\text{rot}} = \frac{\mathbf{w}}{\|\mathbf{w}\|}$ with respect to vector $\mathbf{w}$ yields the orthogonal projection matrix $\mathbf{P}_{\mathbf{w}}^{\perp}$:

$$\frac{\partial \mathbf{n}_{\text{rot}}}{\partial \mathbf{w}} = \frac{1}{\|\mathbf{w}\|} \left(\mathbf{I}_3 - \frac{\mathbf{w} \mathbf{w}^T}{\|\mathbf{w}\|^2} \right) = \frac{1}{\|\mathbf{w}\|} (\mathbf{I}_3 - \mathbf{n}_{\text{rot}} \mathbf{n}_{\text{rot}}^T).$$

Applying the chain rule gives the final composite Jacobian $\mathbf{J}_{i,j}^{\mathbf{n}_{\text{rot}}} = \frac{\partial \mathbf{n}_{\text{rot}}}{\partial \xi_{i,j}} \in \mathbb{R}^{3 \times 6}$:

$$\mathbf{J}_{i,j}^{\mathbf{n}_{\text{rot}}} = \frac{1}{\|\mathbf{w}\|} (\mathbf{I}_3 - \mathbf{n}_{\text{rot}} \mathbf{n}_{\text{rot}}^T) \mathbf{J}_{i,j}^{\mathbf{w}}.$$

This closed-form formulation matches the graph-evaluation structure implemented in CasADi symbolic graphs, enabling rapid parallel evaluation of $\mathcal{J}_{\text{sens}}$ without Monte Carlo sampling loops.

6 Simulation Results

6.1 Visualization

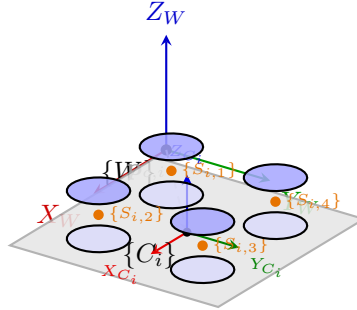


Figure 2: Kinematic spatial layout of cluster frame $\{C_i\}$ and sub-coil frames $\{S_{i,j}\}$.

6.2 Performance Tables

Table 1: Performance comparison across nominal cluster polar tilt angles θ_0 .

Tilt Configuration (θ_0)	Center $\ B\ _{\text{rms}}$	Mean $\mathbf{n}_{\text{rot}}$ Spread	Max $\mathbf{n}_{\text{rot}}$ Spread	Max $\ B\ $ Noise
Outward Tilt (+30.0°)	1.595 mT	0.9348°	2.8114°	4.1701%
Outward Tilt (+15.0°)	3.3465 mT	0.5203°	0.7376°	3.3569%
Flat Baseline (0.0°)	4.2846 mT	0.4128°	0.5846°	2.3216%
Optimal Inward ($\theta_0 = -10.0^\circ$)	4.4294 mT	0.3821°	0.5745°	1.8283%
Optimal Inward ($\theta_0 = -15.0^\circ$)	4.4072 mT	0.3710°	0.5883°	1.7652%
Extreme Inward ($\theta_0 = -30.0^\circ$)	4.1578 mT	0.3352°	0.5967°	2.1985%

Table 2: Comparison of cluster-level tilting vs. individual sub-coil dish focusing.

Architecture	Center $\ B\ _{\text{rms}}$	Mean $\mathbf{n}_{\text{rot}}$ Spread	Max $\mathbf{n}_{\text{rot}}$ Spread	Degrees of Freedom
Outward Cluster ($\theta_0 = +15^\circ$)	3.3465 mT	0.4760°	0.6470°	4
Flat Cluster ($\theta_0 = 0^\circ$)	4.2846 mT	0.4144°	0.5550°	4
Inward Cluster ($\theta_0 = -15^\circ$)	4.4072 mT	0.3710°	0.5883°	4
Individual Sub-Coil Dish	5.4295 mT	0.4121°	0.9396°	16